PROYECTO FINAL DE MAESTRÍA

Análisis y predicción del abandono de clientes de telecomunicaciones

Telco Customer Churn

Objetivo general

Analizar los factores asociados al abandono de clientes de telecomunicaciones y estimar su riesgo según características, servicios, antigüedad y cargos, para orientar estrategias de retención.

Archivos CSV	Registros integrados	Clientes únicos	Abandono observado · por fila
3	7.303	7.043	26,6 %

La tasa inicial incluye filas repetidas. Los modelos utilizan una fila por cliente.

Archivos de datos

Se leen los tres CSV de data/ y se reúnen sus filas en una sola tabla, conservando las mismas 21 columnas.

Archivo	Filas	Columnas
telco_csv_1.csv	2395	21
telco_csv_2.csv	2394	21
telco_csv_3.csv	2514	21

Ver datos de

Integración completa

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte

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telco_csv_1.csv	2395	21
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Ver datos de

Integración completa

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
8064-RAVOH	Male	0	No	Yes	1.0	Yes	No	DSL	No	No	Yes
0310-MVLET	Female	0	Yes	Yes	61.0	Yes	Yes	Fiber optic	No	Yes	No
7714-YXSMB	Female	0	No	No	26.0	Yes	Yes	Fiber optic	No	No	Yes
3675-EQOZA	Male	0	No	No	5.0	Yes	No	No	No internet serv	No internet serv	No internet
1925-TIBLE	Female	0	Yes	No	7.0	Yes	Yes	Fiber optic	No	No	No
2386-OWURY	Female	0	No	No	58.0	Yes	No	Fiber optic	No	No	No
4662-EKDPQ	Male	0	No	No	2.0	Yes	Yes	DSL	No	No	No
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes

Se muestran las primeras 30 filas con los nombres y valores del archivo. Los CSV de entrada no se modifican.

Descargar CSV seleccionado

8064-RAVOH	Male	0	No	Yes	1.0	Yes	No	DSL	No	No	Yes
0310-MVLET	Female	0	Yes	Yes	61.0	Yes	Yes	Fiber optic	No	Yes	No
7714-YXSMB	Female	0	No	No	26.0	Yes	Yes	Fiber optic	No	No	Yes
3675-EQOZA	Male	0	No	No	5.0	Yes	No	No	No internet serv	No internet serv	No internet
1925-TIBLE	Female	0	Yes	No	7.0	Yes	Yes	Fiber optic	No	No	No
2386-OWURY	Female	0	No	No	58.0	Yes	No	Fiber optic	No	No	No
4662-EKDPQ	Male	0	No	No	2.0	Yes	Yes	DSL	No	No	No
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes

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Descargar CSV seleccionado

Detalles de la integración

Los archivos se unen en orden 1 → 2 → 3. Se conserva el archivo y la fila de origen para revisar repeticiones y diferencias entre registros del mismo cliente.

Archivo	Filas	Columnas	SHA-256
telco_csv_1.csv	2395	21	4699d2054e8c740c6899aef71f6e5374fc631c8b2152cecff254f292d775da56
telco_csv_2.csv	2394	21	64ff0b1da5c5ba5d180f3b5169f007c0b37f09f5c3d4045fa3712919d2172e3d
telco_csv_3.csv	2514	21	def8c8739d9c5adc4a9e05efb9cc410f8cd027d7002d672cfc1daa726b9e37f3

Descargar tabla

Diccionario de datos

Variable	Descripción	Tipo

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telco_csv_3.csv	2514	21	def8c8739d9c5adc4a9e05efb9cc410f8cd027d7002d672cfc1daa726b9e37f3

Descargar tabla

Diccionario de datos

Variable	Descripción	Tipo
customerID	Código alfanumérico que identifica a cada cliente y permite detectar registros repetidos.	Categórica
gender	Género registrado del cliente: femenino o masculino.	Categórica
SeniorCitizen	Indica si el cliente está clasificado como adulto mayor: 1 significa sí y 0 significa no.	Categórica
Partner	Indica si el cliente tiene pareja, según la información registrada.	Categórica
Dependents	Indica si el cliente tiene personas dependientes a su cargo.	Categórica
tenure	Cantidad de meses que el cliente ha permanecido en la empresa de telecomunicaciones.	Numérica
PhoneService	Indica si el cliente tiene contratado el servicio de telefonía.	Categórica
MultipleLines	Indica si el cliente dispone de varias líneas telefónicas o no tiene servicio de telefonía.	Categórica
InternetService	Tipo de servicio de internet contratado: DSL, fibra óptica o sin servicio de internet.	Categórica

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tenure	Cantidad de meses que el cliente ha permanecido en la empresa de telecomunicaciones.	Numérica
PhoneService	Indica si el cliente tiene contratado el servicio de telefonía.	Categórica
MultipleLines	Indica si el cliente dispone de varias líneas telefónicas o no tiene servicio de telefonía.	Categórica
InternetService	Tipo de servicio de internet contratado: DSL, fibra óptica o sin servicio de internet.	Categórica
OnlineSecurity	Indica si el cliente tiene seguridad en línea, no la tiene o no dispone de internet.	Categórica
OnlineBackup	Indica si el cliente tiene respaldo en línea, no lo tiene o no dispone de internet.	Categórica
DeviceProtection	Indica si el cliente tiene protección de dispositivos, no la tiene o no dispone de internet.	Categórica
TechSupport	Indica si el cliente tiene soporte técnico contratado, no lo tiene o no dispone de internet.	Categórica
StreamingTV	Indica si el cliente tiene servicio de televisión por streaming o no dispone de este servicio o de internet.	Categórica
StreamingMovies	Indica si el cliente tiene servicio de películas por streaming o no dispone de este servicio o de internet.	Categórica
Contract	Duración del contrato del cliente: mensual, de un año o de dos años.	Categórica
PaperlessBilling	Indica si el cliente utiliza facturación electrónica, sin recibir una factura en papel.	Categórica

OnlineSecurity	Indica si el cliente tiene seguridad en línea, no la tiene o no dispone de internet.	Categórica
OnlineBackup	Indica si el cliente tiene respaldo en línea, no lo tiene o no dispone de internet.	Categórica
DeviceProtection	Indica si el cliente tiene protección de dispositivos, no la tiene o no dispone de internet.	Categórica
TechSupport	Indica si el cliente tiene soporte técnico contratado, no lo tiene o no dispone de internet.	Categórica
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StreamingMovies	Indica si el cliente tiene servicio de películas por streaming o no dispone de este servicio o de internet.	Categórica
Contract	Duración del contrato del cliente: mensual, de un año o de dos años.	Categórica
PaperlessBilling	Indica si el cliente utiliza facturación electrónica, sin recibir una factura en papel.	Categórica
PaymentMethod	Método de pago registrado: cheque electrónico, cheque por correo, transferencia automática o tarjeta automática.	Categórica
MonthlyCharges	Importe mensual cobrado al cliente por sus servicios, expresado en unidades monetarias.	Numérica
TotalCharges	Importe total acumulado cobrado al cliente durante su permanencia, expresado en unidades monetarias.	Numérica
Churn	Indica si el cliente abandonó el servicio: Yes corresponde a abandono y No a permanencia.	Categórica

Fuente original en Kaggle

Proyecto académico · Natalia Gutiérrez Soliz · Datos locales

Integración completa · tabla 1 · filas 1, columnas 1

Telco · Inicio · 22/10/2026

Archivo	Filas	Columnas
telco_csv_1.csv	2395	21
telco_csv_2.csv	2394	21
telco_csv_3.csv	2514	21

Integración completa · tabla 2 · filas 1, columnas 1

Telco · Inicio · 22/10/2026

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
8064-RAVOH	Male	0	No	Yes	1.0	Yes	No	DSL	No	No	Yes
0310-MVLET	Female	0	Yes	Yes	61.0	Yes	Yes	Fiber optic	No	Yes	No
7714-YXSMB	Female	0	No	No	26.0	Yes	Yes	Fiber optic	No	No	Yes
3675-EQOZA	Male	0	No	No	5.0	Yes	No	No	No internet serv	No internet serv	No internet
1925-TIBLE	Female	0	Yes	No	7.0	Yes	Yes	Fiber optic	No	No	No
2386-OWURY	Female	0	No	No	58.0	Yes	No	Fiber optic	No	No	No
4662-EKDPQ	Male	0	No	No	2.0	Yes	Yes	DSL	No	No	No
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes

Integración completa · tabla 2 · filas 1, columnas 2

Telco · Inicio · 22/10/2026

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	No	DSL	No	No	Yes	No	No	No	Month-to-month	Yes
	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	No	No	No	Yes	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	Yes	Yes	Yes	Month-to-month	Yes
	Yes	DSL	No	No	No	No	No	Yes	Month-to-month	No
	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Two year	Yes
	No	DSL	No	No	No	Yes	No	No	Two year	No
	No	DSL	No	No	Yes	No	Yes	No	Month-to-month	Yes

Integración completa · tabla 2 · filas 1, columnas 3

Telco · Inicio · 22/10/2026

eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	Yes	Electronic check	49.85	49.85	No
	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.15	6010.05	Yes
	No	Yes	Yes	Month-to-month	Yes	Bank transfer (automatic)	100.5	2599.95	No
ternet service	No internet se	No internet se	No internet service	Month-to-month	No	Bank transfer (automatic)	20.65		No
	No	Yes	No	Month-to-month	Yes	Electronic check	86.5	582.5	Yes
	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	95.3	5817.7	No
	No	No	Yes	Month-to-month	No	Bank transfer (automatic)	62.05	118.3	Yes
	Yes	Yes	Yes	Two year	Yes	Bank transfer (automatic)	106.5	7348.8	Yes
	Yes	No	No	Two year	No	Bank transfer (automatic)	50.55	3260.1	No
	No	Yes	No	Month-to-month	Yes	Bank transfer (automatic)	60.05	1236.15	Yes

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes
8042-RNLKO	Male	0	No	No	66.0	Yes	Yes	No	No internet serv	No internet serv	No internet
4119-ZYPZY	Male	1	No	No	71.0	Yes	Yes	Fiber optic	No	Yes	No
4800-VHZKI	Female	0	No	No	1.0	Yes	No	No	No internet serv	No internet serv	No internet
6941-PMGEP	Female	0	No	No	5.0	Yes	No	Fiber optic	No	No	No
1410-RSCMR	Male	0	Yes	Yes	7.0	Yes	No	DSL	Yes	No	Yes
0670-ANMUU	Male	0	No	No	71.0	Yes	Yes	Fiber optic	Yes	No	No
1024-GUALD	Female	0	Yes	No		No	No phone servi	DSL	No	No	No

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Two year	Yes
	No	DSL	No	No	No	Yes	No	No	Two year	No
	No	DSL	No	No	Yes	No	Yes	No	Month-to-month	Yes
	Yes	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	One year	No
	Yes	Fiber optic	No	Yes	No	Yes	Yes	Yes		Yes
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	Yes
	No	Fiber optic	No	No	No	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	Yes	No	No	Yes	No	Yes	One year	No
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	Yes

eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	Yes	Yes	Yes	Two year	Yes	Bank transfer (automatic)	106.5	7348.8	Yes
	Yes	No	No	Two year	No	Bank transfer (automatic)	50.55	3260.1	No
	No	Yes	No	Month-to-month	Yes	Bank transfer (automatic)	60.05	1236.15	Yes
ternet service	No internet ser	No internet se	No internet service	One year	No	Bank transfer (automatic)	25.45	1699.15	No
	Yes	Yes	Yes		Yes	Electronic check	106.8	7623.2	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	Yes	Mailed check	19.9		Yes
	No	No	Yes	Month-to-month	Yes	Electronic check	80.0	412.5	Yes
	Yes	No	Yes	Month-to-month	Yes	Credit card (automatic)	71.35		No
	Yes	No	Yes	One year	No	Credit card (automatic)		6687.85	No
	No	No	No	Month-to-month	Yes	Electronic check	24.8	24.8	Yes

Integración completa · tabla 2 · filas 3, columnas 1

Telco · Inicio · 22/10/2026

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
6941-PMGEP	Female	0	No	No	5.0	Yes	No	Fiber optic	No	No	No
1410-RSCMR	Male	0	Yes	Yes	7.0	Yes	No	DSL	Yes	No	Yes
0670-ANMUU	Male	0	No	No	71.0	Yes	Yes	Fiber optic	Yes	No	No
1024-GUALD	Female	0	Yes	No		No	No phone servi	DSL	No	No	No
1640-PLFMP	Female	0	No	No	1.0	Yes	No	Fiber optic	No	No	No
3771-PZOBW	Male	0	No	No	20.0	Yes	Yes	Fiber optic	No	No	Yes
9371-BITHB	Female	0	Yes	Yes	52.0	Yes	No	No	No internet serv	No internet serv	No internet
7564-GHCVB	Female	0	Yes	Yes	72.0	Yes	Yes	Fiber optic	Yes	No	No
3679-XASPY	Female	0	Yes	Yes	1.0	Yes	No	No	No internet serv	No internet serv	No internet
5858-EAFCZ	Male	0	No	No	1.0	Yes	No	DSL	No	No	No

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	No	Fiber optic	No	No	No	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	Yes	No	No	Yes	No	Yes	One year	No
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	Yes	No		Yes
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Two year	No
	Yes	Fiber optic	Yes	No	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	No
	No	DSL	No	No	No	No	No	No	Month-to-month	No
	No	DSL	Yes	No	Yes	No	No	Yes	One year	No

eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	Yes	Month-to-month	Yes	Electronic check	80.0	412.5	Yes
	Yes	No	Yes	Month-to-month	Yes	Credit card (automatic)	71.35		No
	Yes	No	Yes	One year	No	Credit card (automatic)		6687.85	No
	No	No	No	Month-to-month	Yes	Electronic check	24.8	24.8	Yes
	No	No	No	Month-to-month	Yes	Electronic check	70.25	70.25	No
	No	Yes	No		Yes	Credit card (automatic)	90.7	1781.35	No
ternet service	No internet se	No internet se	No internet service	Two year	No	Mailed check	19.85	1070.5	No
	No	No	No	Two year	No	Bank transfer (automatic)	79.05		No
ternet service	No internet se	No internet se	No internet service	Month-to-month	No	Electronic check	19.45	19.45	No
	No	No	No	Month-to-month	No	Mailed check	44.45	44.45	No

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
7564-GHCVB	Female	0	Yes	Yes	72.0	Yes	Yes	Fiber optic	Yes	No	No
3679-XASPY	Female	0	Yes	Yes	1.0	Yes	No	No	No internet serv	No internet serv	No internet
5858-EAFCZ	Male	0	No	No	1.0	Yes	No	DSL	No	No	No
4223-BKEOR	Female	0	No	Yes	21.0	Yes	No	DSL	Yes	No	Yes
4770-QAZXN	Female	0	No	No	13.0	Yes	No	No	No internet serv	No internet serv	No internet
1470-PSXNM	Male	0	Yes	Yes	49.0	Yes	Yes	Fiber optic	No	Yes	Yes
0564-JJHGS	Male	0	No	No	9.0	Yes	Yes	Fiber optic	No	No	No
4695-WJZUE	Female	1	No	No	2.0	Yes	Yes	Fiber optic	No	No	Yes
8200-KLNYW	Female	0	Yes	No	21.0	Yes	No	DSL	Yes	No	Yes
4396-KLSEH	Male	0	No	No	4.0	Yes	No	No	No internet serv	No internet serv	No internet

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	Yes	Fiber optic	Yes	No	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	No
	No	DSL	No	No	No	No	No	No	Month-to-month	No
	No	DSL	Yes	No	Yes	No	No	Yes	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	Yes	Yes	No	Yes	No	One year	Yes
	Yes	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	Yes	Yes	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	Yes

eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Two year	No	Bank transfer (automatic)	79.05		No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Electronic check	19.45	19.45	No
	No	No	No	Month-to-month	No	Mailed check	44.45	44.45	No
	No	No	Yes	One year	No	Mailed check	64.85	1336.8	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Credit card (automatic)	19.45	232.1	No
	No	Yes	No	One year	Yes	Electronic check	93.65	4520.15	No
	No	Yes	Yes	Month-to-month	Yes	Electronic check	95.5	829.1	Yes
	No	No	Yes	Month-to-month	Yes	Electronic check	88.55		Yes
	Yes	Yes	Yes	One year	No	Mailed check	80.9	1714.95	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	Yes	Electronic check	19.85	63.0	No

Integración completa · tabla 3 · filas 1, columnas 1

Telco · Inicio · 22/10/2026

Archivo	Filas	Columnas	SHA-256
telco_csv_1.csv	2395	21	4699d2054e8c740c6899aef71f6e5374fc631c8b2152cecff254f292d775da56
telco_csv_2.csv	2394	21	64ff0b1da5c5ba5d180f3b5169f007c0b37f09f5c3d4045fa3712919d2172e3d
telco_csv_3.csv	2514	21	def8c8739d9c5adc4a9e05efb9cc410f8cd027d7002d672cfc1daa726b9e37f3

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 1, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
8064-RAVOH	Male	0	No	Yes	1.0	Yes	No	DSL	No	No	Yes
0310-MVLET	Female	0	Yes	Yes	61.0	Yes	Yes	Fiber optic	No	Yes	No
7714-YXSMB	Female	0	No	No	26.0	Yes	Yes	Fiber optic	No	No	Yes
3675-EQOZA	Male	0	No	No	5.0	Yes	No	No	No internet serv	No internet serv	No internet
1925-TIBLE	Female	0	Yes	No	7.0	Yes	Yes	Fiber optic	No	No	No
2386-OWURY	Female	0	No	No	58.0	Yes	No	Fiber optic	No	No	No
4662-EKDPQ	Male	0	No	No	2.0	Yes	Yes	DSL	No	No	No
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 1, columnas 2

Telco · Inicio · 22/10/2026

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	No	DSL	No	No	Yes	No	No	No	Month-to-month	Yes
	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	No	No	No	Yes	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	Yes	Yes	Yes	Month-to-month	Yes
	Yes	DSL	No	No	No	No	No	Yes	Month-to-month	No
	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Two year	Yes
	No	DSL	No	No	No	Yes	No	No	Two year	No
	No	DSL	No	No	Yes	No	Yes	No	Month-to-month	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 1, columnas 3

Telco · Inicio · 22/10/2026

eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	Yes	Electronic check	49.85	49.85	No
	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.15	6010.05	Yes
	No	Yes	Yes	Month-to-month	Yes	Bank transfer (automatic)	100.5	2599.95	No
ternet service	No internet se	No internet se	No internet service	Month-to-month	No	Bank transfer (automatic)	20.65		No
	No	Yes	No	Month-to-month	Yes	Electronic check	86.5	582.5	Yes
	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	95.3	5817.7	No
	No	No	Yes	Month-to-month	No	Bank transfer (automatic)	62.05	118.3	Yes
	Yes	Yes	Yes	Two year	Yes	Bank transfer (automatic)	106.5	7348.8	Yes
	Yes	No	No	Two year	No	Bank transfer (automatic)	50.55	3260.1	No
	No	Yes	No	Month-to-month	Yes	Bank transfer (automatic)	60.05	1236.15	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 2, columnas 1

Telco · Inicio · 22/10/2026

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
5748-RNCJT	Male	0	No	No	69.0	Yes	Yes	Fiber optic	No	No	Yes
8587-XYZSF	Male	0	No	No	67.0	Yes	No	DSL	No	No	No
8399-YNDCH	Male	1	No	No	21.0	Yes	No	DSL	No	No	Yes
8042-RNLKO	Male	0	No	No	66.0	Yes	Yes	No	No internet serv	No internet serv	No internet
4119-ZYPZY	Male	1	No	No	71.0	Yes	Yes	Fiber optic	No	Yes	No
4800-VHZKI	Female	0	No	No	1.0	Yes	No	No	No internet serv	No internet serv	No internet
6941-PMGEP	Female	0	No	No	5.0	Yes	No	Fiber optic	No	No	No
1410-RSCMR	Male	0	Yes	Yes	7.0	Yes	No	DSL	Yes	No	Yes
0670-ANMUU	Male	0	No	No	71.0	Yes	Yes	Fiber optic	Yes	No	No
1024-GUALD	Female	0	Yes	No		No	No phone servi	DSL	No	No	No

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 2, columnas 2

Telco · Inicio · 22/10/2026

Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Two year	Yes
	No	DSL	No	No	No	Yes	No	No	Two year	No
	No	DSL	No	No	Yes	No	Yes	No	Month-to-month	Yes
	Yes	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	No
	Yes	Fiber optic	No	Yes	No	Yes	Yes	Yes		Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	Yes
	No	Fiber optic	No	No	No	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	Yes	No	No	Yes	No	Yes	One year	No
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 2, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	Yes	Yes	Yes	Two year	Yes	Bank transfer (automatic)	106.5	7348.8	Yes
	Yes	No	No	Two year	No	Bank transfer (automatic)	50.55	3260.1	No
	No	Yes	No	Month-to-month	Yes	Bank transfer (automatic)	60.05	1236.15	Yes
ternet service	No internet ser	No internet se	No internet service	One year	No	Bank transfer (automatic)	25.45	1699.15	No
	Yes	Yes	Yes		Yes	Electronic check	106.8	7623.2	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	Yes	Mailed check	19.9		Yes
	No	No	Yes	Month-to-month	Yes	Electronic check	80.0	412.5	Yes
	Yes	No	Yes	Month-to-month	Yes	Credit card (automatic)	71.35		No
	Yes	No	Yes	One year	No	Credit card (automatic)		6687.85	No
	No	No	No	Month-to-month	Yes	Electronic check	24.8	24.8	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 3, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
6941-PMGEP	Female	0	No	No	5.0	Yes	No	Fiber optic	No	No	No
1410-RSCMR	Male	0	Yes	Yes	7.0	Yes	No	DSL	Yes	No	Yes
0670-ANMUU	Male	0	No	No	71.0	Yes	Yes	Fiber optic	Yes	No	No
1024-GUALD	Female	0	Yes	No		No	No phone servi	DSL	No	No	No
1640-PLFMP	Female	0	No	No	1.0	Yes	No	Fiber optic	No	No	No
3771-PZOBW	Male	0	No	No	20.0	Yes	Yes	Fiber optic	No	No	Yes
9371-BITHB	Female	0	Yes	Yes	52.0	Yes	No	No	No internet serv	No internet serv	No internet
7564-GHCVB	Female	0	Yes	Yes	72.0	Yes	Yes	Fiber optic	Yes	No	No
3679-XASPY	Female	0	Yes	Yes	1.0	Yes	No	No	No internet serv	No internet serv	No internet
5858-EAFCZ	Male	0	No	No	1.0	Yes	No	DSL	No	No	No
4223-BKEQP	Female	0	No	Yes	21.0	Yes	No	DSL	Yes	No	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 3, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	No	Fiber optic	No	No	No	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	Yes	No	No	Yes	No	Yes	One year	No
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	Yes	No		Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	Yes	Fiber optic	Yes	No	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	No	DSL	No	No	No	No	No	No	Month-to-month	No
	No	DSL	Yes	No	Yes	No	No	Yes	One year	No

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 3, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	Yes	Month-to-month	Yes	Electronic check	80.0	412.5	Yes
	Yes	No	Yes	Month-to-month	Yes	Credit card (automatic)	71.35		No
	Yes	No	Yes	One year	No	Credit card (automatic)		6687.85	No
	No	No	No	Month-to-month	Yes	Electronic check	24.8	24.8	Yes
	No	No	No	Month-to-month	Yes	Electronic check	70.25	70.25	No
	No	Yes	No		Yes	Credit card (automatic)	90.7	1781.35	No
ternet service	No internet se	No internet se	No internet service	Two year	No	Mailed check	19.85	1070.5	No
	No	No	No	Two year	No	Bank transfer (automatic)	79.05		No
ternet service	No internet se	No internet se	No internet service	Month-to-month	No	Electronic check	19.45	19.45	No
	No	No	No	Month-to-month	No	Mailed check	44.45	44.45	No

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 4, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
7564-GHCVB	Female	0	Yes	Yes	72.0	Yes	Yes	Fiber optic	Yes	No	No
3679-XASPY	Female	0	Yes	Yes	1.0	Yes	No	No	No internet serv	No internet serv	No internet
5858-EAFCZ	Male	0	No	No	1.0	Yes	No	DSL	No	No	No
4223-BKEOR	Female	0	No	Yes	21.0	Yes	No	DSL	Yes	No	Yes
4770-QAZXN	Female	0	No	No	13.0	Yes	No	No	No internet serv	No internet serv	No internet
1470-PSXNM	Male	0	Yes	Yes	49.0	Yes	Yes	Fiber optic	No	Yes	Yes
0564-JJHGS	Male	0	No	No	9.0	Yes	Yes	Fiber optic	No	No	No
4695-WJZUE	Female	1	No	No	2.0	Yes	Yes	Fiber optic	No	No	Yes
8200-KLNYW	Female	0	Yes	No	21.0	Yes	No	DSL	Yes	No	Yes
4396-KLSEH	Male	0	No	No	4.0	Yes	No	No	No internet serv	No internet serv	No internet

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 4, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	Yes	Fiber optic	Yes	No	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	No	DSL	No	No	No	No	No	No	Month-to-month	No
	No	DSL	Yes	No	Yes	No	No	Yes	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	Yes	Yes	No	Yes	No	One year	Yes
	Yes	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	No	Yes	No	No	Yes	Month-to-month	Yes
	No	DSL	Yes	No	Yes	Yes	Yes	Yes	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	Yes

Vista previa: telco_csv_1.csv · primeras 30 filas · filas 4, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Two year	No	Bank transfer (automatic)	79.05		No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Electronic check	19.45	19.45	No
	No	No	No	Month-to-month	No	Mailed check	44.45	44.45	No
	No	No	Yes	One year	No	Mailed check	64.85	1336.8	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Credit card (automatic)	19.45	232.1	No
	No	Yes	No	One year	Yes	Electronic check	93.65	4520.15	No
	No	Yes	Yes	Month-to-month	Yes	Electronic check	95.5	829.1	Yes
	No	No	Yes	Month-to-month	Yes	Electronic check	88.55		Yes
	Yes	Yes	Yes	One year	No	Mailed check	80.9	1714.95	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	Yes	Electronic check	19.85	63.0	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 1, columnas 1

Telco · Inicio · 22/10/2026

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
9907-SWKKF	Female	1	No	No	1.0	No	No phone servi	DSL	No	No	No
1843-TLSGD	Female	0	Yes	Yes	13.0	Yes	No	No	No internet serv	No internet serv	No internet
6108-OQZDQ	Female	0	Yes	Yes	4.0	Yes	Yes	Fiber optic	No	Yes	No
2660-EMUBI	Male	1	No	No	10.0	Yes	Yes	Fiber optic	No	No	No
2275-RBYQS	Female	0	No	No	1.0	Yes	No	DSL	No	No	No
9481-IEBZY	Male	1	Yes	No	72.0	Yes	Yes	Fiber optic	Yes	Yes	Yes
4393-RYCRE	Male	0	No	No	44.0	Yes	No	Fiber optic	Yes	Yes	Yes
7619-ODSGN	Male	0	Yes	Yes	24.0	Yes	Yes	DSL	Yes	No	No
6692-YQHXC	Male	0	No	No	16.0	Yes	No	Fiber optic	No	No	No
0384-RVBPI	Male	0	No	No	20.0	Yes	No	DSL	Yes	No	Yes

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 1, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes
	No	DSL	No	No	No	No	No	No	Month-to-month	Yes
	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Month-to-month	Yes
	No	Fiber optic	Yes	Yes	Yes	No	Yes	Yes	One year	Yes
	Yes	DSL	Yes	No	No	No	No	No	Month-to-month	No
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	No	DSL	Yes	No	Yes	No	No	Yes	Month-to-month	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 1, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	No	Mailed check	25.05	25.05	Yes
ternet service	No internet ser	No internet se	No internet service	Two year	No	Mailed check	20.85	272.35	No
	No	Yes	Yes	Month-to-month	Yes	Electronic check	-99.0	384.5	Yes
	No	Yes	Yes	Month-to-month	Yes	Electronic check	-99.0	898.35	Yes
	No	No	No	Month-to-month	Yes	Mailed check	45.4	45.4	No
	Yes	Yes	Yes	Month-to-month	Yes	Credit card (automatic)	112.9	8061.5	No
	No	Yes	Yes	One year	Yes	Electronic check	106.05	4510.8	No
	No	No	No	Month-to-month	No	Electronic check	54.75	1338.15	Yes
	No	No	No	Month-to-month	Yes	Credit card (automatic)	69.95	1205.5	No
	No	No	Yes	Month-to-month	No	Credit card (automatic)	64.4	1398.6	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 2, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
7619-ODSGN	Male	0	Yes	Yes	24.0	Yes	Yes	DSL	Yes	No	No
6692-YQHXC	Male	0	No	No	16.0	Yes	No	Fiber optic	No	No	No
0384-RVBPI	Male	0	No	No	20.0	Yes	No	DSL	Yes	No	Yes
7797-EJMDP	Female	0	No	No	8.0	Yes	No	DSL	Yes	No	No
5529-GIBVH	Female	0	No	No	47.0	Yes	No	No	No internet serv	No internet serv	No internet
7322-OCWHC	Male	1	Yes	No	72.0	Yes	Yes	Fiber optic	Yes	Yes	Yes
8735-IJJEG	Male	0	Yes	No	14.0	Yes	No	DSL	No	No	Yes
2346-CZYIL	Male	0	No	No	27.0	Yes	No	No	No internet serv	No internet serv	No internet
4846-WHAFZ	Female	1	Yes	No	37.0	Yes	Yes	Fiber optic	No	No	No
9114-DPSIA	Male	0	Yes	Yes	72.0	Yes	Yes	DSL	Yes	Yes	Yes

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 2, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	Yes	DSL	Yes	No	No	No	No	No	Month-to-month	No
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	No	DSL	Yes	No	Yes	No	No	Yes	Month-to-month	No
	No	DSL	Yes	No	No	Yes	No	Yes	Month-to-month	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	Yes	Fiber optic	Yes	Yes	Yes	Yes	No	No	Two year	Yes
	No	DSL	No	No	Yes	No	No	No	Month-to-month	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	Yes	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	Yes	No	Yes	Two year	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 2, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	No	Electronic check	54.75	1338.15	Yes
	No	No	No	Month-to-month	Yes	Credit card (automatic)	69.95	1205.5	No
	No	No	Yes	Month-to-month	No	Credit card (automatic)	64.4	1398.6	No
	Yes	No	Yes	Month-to-month	No	Bank transfer (automatic)	66.05	574.5	No
ternet service	No internet ser	No internet se	No internet service	Two year	No	Mailed check	-99.0	1046.2	No
	Yes	No	No	Two year	Yes	Bank transfer (automatic)	93.05	6735.05	No
	No	No	No	Month-to-month	No	Credit card (automatic)	50.2	668.85	No
ternet service	No internet ser	No internet se	No internet service	Two year	No	Credit card (automatic)	20.35	531.6	No
	No	No	No	Month-to-month	Yes	Electronic check	76.5	2868.15	Yes
	Yes	No	Yes	Two year	No	Credit card (automatic)	81.0	5750.0	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 3, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
8735-IJJEG	Male	0	Yes	No	14.0	Yes	No	DSL	No	No	Yes
2346-CZYIL	Male	0	No	No	27.0	Yes	No	No	No internet serv	No internet serv	No internet
4846-WHAFZ	Female	1	Yes	No	37.0	Yes	Yes	Fiber optic	No	No	No
9114-DPSIA	Male	0	Yes	Yes	72.0	Yes	Yes	DSL	Yes	Yes	Yes
8485-GJCDN	Female	1	No	No	5.0	Yes	No	Fiber optic	No	No	No
8008-HAWED	Male	0	No	No	14.0	No	No phone servi	DSL	No	No	Yes
7218-HKQFK	Male	0	Yes	No	2.0	Yes	No	Fiber optic	No	No	No
9483-GCPWE	Male	0	No	Yes	9.0	Yes	No	No	No internet serv	No internet serv	No internet
1088-CNNKB	Male	0	Yes	No	63.0	Yes	Yes	DSL	Yes	Yes	Yes
0541-FITGH	Female	0	Yes	No	2.0	Yes	No	DSL	Yes	No	No
[illegible]	[illegible]	0	No	No	1.0	Yes	No	Fiber optic	Yes	Yes	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 3, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	No	DSL	No	No	Yes	No	No	No	Month-to-month	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	Yes	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	Yes	No	Yes	Two year	No
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	No phone servi	DSL	No	No	Yes	No	Yes	Yes	Month-to-month	Yes
	No	Fiber optic	No	No	No	Yes	Yes	Yes	Month-to-month	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	No
	Yes	DSL	Yes	Yes	Yes	Yes	No	Yes	One year	No
	No	DSL	Yes	No	No	No	No	Yes	Month-to-month	Yes
	No	Fiber optic	Yes	Yes	No	Yes	No	No	Month-to-month	Yes

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 3, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	No	Credit card (automatic)	50.2	668.85	No
ternet service	No internet ser	No internet se	No internet service	Two year	No	Credit card (automatic)	20.35	531.6	No
	No	No	No	Month-to-month	Yes	Electronic check	76.5	2868.15	Yes
	Yes	No	Yes	Two year	No	Credit card (automatic)	81.0	5750.0	No
	No	No	No	Month-to-month	Yes	Mailed check	69.05	318.5	Yes
	No	Yes	Yes	Month-to-month	Yes	Electronic check	50.75	712.75	Yes
	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	94.2	167.5	Yes
ternet service	No internet ser	No internet se	No internet service	One year	No	Mailed check	20.1	190.25	No
	Yes	No	Yes	One year	No	Credit card (automatic)	-99.0	5040.2	No
	No	No	Yes	Month-to-month	Yes	Mailed check	62.15	113.1	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 4, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
9483-GCPWE	Male	0	No	Yes	9.0	Yes	No	No	No internet serv	No internet serv	No internet
1088-CNNKB	Male	0	Yes	No	63.0	Yes	Yes	DSL	Yes	Yes	Yes
0541-FITGH	Female	0	Yes	No	2.0	Yes	No	DSL	Yes	No	No
2792-LSHWX	Female	0	No	No	1.0	Yes	No	Fiber optic	Yes	Yes	No
0963-ZBDRN	Male	0	No	No	32.0	Yes	No	Fiber optic	No	No	No
9717-IOAAF	Male	0	No	Yes	27.0	Yes	Yes	No	No internet serv	No internet serv	No internet
9003-CPATH	Male	0	No	No	42.0	Yes	Yes	Fiber optic	No	No	Yes
0513-RBGPE	Male	0	Yes	Yes	37.0	Yes	Yes	DSL	Yes	No	Yes
4664-NJCMS	Female	0	Yes	No	33.0	No	No phone servi	DSL	Yes	No	Yes
8591-TKMZH	Male	0	Yes	Yes	59.0	Yes	Yes	Fiber optic	Yes	Yes	No

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 4, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	No
	Yes	DSL	Yes	Yes	Yes	Yes	No	Yes	One year	No
	No	DSL	Yes	No	No	No	No	Yes	Month-to-month	Yes
	No	Fiber optic	Yes	Yes	No	Yes	No	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	Yes	Yes	Yes	Month-to-month	Yes
	Yes	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	two years	Yes
	Yes	Fiber optic	No	No	Yes	No	No	No	Month-to-month	No
	Yes	DSL	Yes	No	Yes	Yes	No	No	Two year	Yes
	No phone servi	DSL	Yes	No	Yes	Yes	Yes	Yes	Two year	No
	Yes	Fiber optic	Yes	Yes	No	Yes	Yes	Yes	Two year	Yes

Vista previa: telco_csv_2.csv · primeras 30 filas · filas 4, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
ternet service	No internet ser	No internet se	No internet service	One year	No	Mailed check	20.1	190.25	No
	Yes	No	Yes	One year	No	Credit card (automatic)	-99.0	5040.2	No
	No	No	Yes	Month-to-month	Yes	Mailed check	62.15	113.1	No
	Yes	No	No	Month-to-month	Yes	Mailed check	83.4	83.4	No
	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	96.2	3183.4	Yes
ternet service	No internet ser	No internet se	No internet service	two years	Yes	Credit card (automatic)	25.85	788.55	No
	No	No	No	Month-to-month	No	Electronic check	79.9	3313.4	No
	Yes	No	No	Two year	Yes	Credit card (automatic)	62.8	2278.75	No
	Yes	Yes	Yes	Two year	No	Bank transfer (automatic)	59.55		No
	Yes	Yes	Yes	Two year	Yes	Mailed check	111.1	6555.2	No

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 1, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
2672-OJQZP	Female	0	No	No	1.0	Yes	No	DSL	No	No	No
8042-JVNFH	Male	0	No	No	1.0	Yes	No	Fiber optic	No	No	No
6437-UDQJM	Female	1	Yes	No	72.0	Yes	Yes	DSL	Yes	Yes	Yes
0277-ORXQS	Male	0	Yes	Yes	66.0	No	No phone servi	DSL	No	No	No
9699-UBQFS	Female	1	Yes	No	43.0	Yes	Yes	Fiber optic	Yes	Yes	No
2453-SAFNS	Female	1	No	No	54.0	Yes	Yes	DSL	No	Yes	No
4480-MBMLB	Female	0	No	No	36.0	Yes	Yes	Fiber optic	No	Yes	Yes
1599-EAHXY	Male	0	Yes	Yes	60.0	Yes	Yes	No	No internet serv	No internet serv	No internet
9770-LXDBK	Female	0	No	No	3.0	Yes	No	No	No internet serv	No internet serv	No internet
2911-WDXMV	Male	0	No	Yes	18.0	Yes	Yes	Fiber optic	No	No	Yes

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 1, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	No	DSL	No	No	No	No	No	No	Month-to-month	No
	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	No	Yes	Yes	Two year	Yes
	No phone servi	DSL	No	No	No	Yes	Yes	Yes	Two year	No
	Yes	Fiber optic	Yes	Yes	No	No	Yes	No	Month-to-month	Yes
	Yes	DSL	No	Yes	No	Yes	Yes	No	One year	No
	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	One year	Yes
	Yes	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	No	Yes	No	No	No	Month-to-month	Yes

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 1, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	No	Mailed check	44.75	44.75	No
	No	No	No	Month-to-month	Yes	Mailed check	71.35	71.35	Yes
	No	Yes	Yes	Two year	Yes	Credit card (automatic)	84.1	6129.65	No
	Yes	Yes	Yes	Two year	No	Bank transfer (automatic)	50.55	3364.55	No
	No	Yes	No	Month-to-month	Yes	Electronic check	94.5	4156.8	No
	Yes	Yes	No	One year	No	Mailed check	72.1	3886.05	No
	No	Yes	Yes	One year	Yes	Bank transfer (automatic)	104.8	3886.45	No
ternet service	No internet ser	No internet se	No internet service	One year	No	Credit card (automatic)	982.29948722349	1506.4	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Mailed check	20.4	63.15	No
	No	No	No	Month-to-month	Yes	Credit card (automatic)	80.55	1406.65	No

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 2, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
1599-EAHXY	Male	0	Yes	Yes	60.0	Yes	Yes	No	No internet serv	No internet serv	No internet
9770-LXDBK	Female	0	No	No	3.0	Yes	No	No	No internet serv	No internet serv	No internet
2911-WDXMV	Male	0	No	Yes	18.0	Yes	Yes	Fiber optic	No	No	Yes
1775-KWJKQ	Male	0	Yes	Yes	72.0	Yes	Yes	DSL	Yes	Yes	Yes
8622-ZLFKO	Female	0	Yes	No	6.0	Yes	Yes	Fiber optic	No	Yes	No
8166-ORCHU	Male	1	Yes	No	33.0	Yes	No	Fiber optic	Yes	Yes	No
2856-NNASM	Male	1	No	No	43.0	Yes	No	Fiber optic	No	No	No
3714-XPXBW	Female	0	No	No	37.0	Yes	Yes	DSL	Yes	Yes	Yes
7921-BEPCI	Female	0	No	No	72.0	Yes	Yes	DSL	Yes	Yes	No
8739-QOTTN	Female	0	Yes	No	2.0	Yes	No	No	No internet serv	No internet serv	No internet

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 2, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	Yes	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	No	Yes	No	No	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No
	Yes	Fiber optic	No	Yes	No	No	No	Yes	Month-to-month	Yes
	No	Fiber optic	Yes	Yes	No	Yes	No	Yes	One year	Yes
	No	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	Yes	Yes	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 2, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
ternet service	No internet ser	No internet se	No internet service	One year	No	Credit card (automatic)	982.29948722349	1506.4	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Mailed check	20.4	63.15	No
	No	No	No	Month-to-month	Yes	Credit card (automatic)	80.55	1406.65	No
	Yes	Yes	Yes	Two year	No	Credit card (automatic)		6562.9	No
	No	No	Yes	Month-to-month	Yes	Electronic check	90.75	512.25	No
	Yes	No	Yes	One year	Yes	Electronic check	93.55	3055.5	No
	No	Yes	Yes	Month-to-month	Yes	Mailed check	89.55	3856.75	Yes
	Yes	Yes	No	Month-to-month	Yes	Credit card (automatic)		2976.95	No
	No	No	No	Two year	No	Bank transfer (automatic)	59.75	4265.0	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Mailed check	20.35		No

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 3, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
2856-NNASM	Male	1	No	No	43.0	Yes	No	Fiber optic	No	No	No
3714-XPXBW	Female	0	No	No	37.0	Yes	Yes	DSL	Yes	Yes	Yes
7921-BEPCI	Female	0	No	No	72.0	Yes	Yes	DSL	Yes	Yes	No
8739-QOTTN	Female	0	Yes	No	2.0	Yes	No	No	No internet serv	No internet serv	No internet
9121-PHQSR	Male	1	Yes	No	16.0	Yes	Yes	Fiber optic	No	No	No
4987-GQWPO	Male	0	No	No	17.0	No	No phone servi	DSL	No	No	No
6479-VDGRK	Female	0	Yes	Yes	72.0	Yes	No	DSL	Yes	Yes	Yes
8104-OSKWT	Female	0	No	No	12.0	Yes	Yes	Fiber optic	Yes	No	No
3688-FTHLT	Female	0	No	No	16.0	Yes	Yes	DSL	No	No	No
2034-CGRHZ	Male	1	No	No	24.0	Yes	Yes	Fiber optic	No	Yes	Yes
2667-WYUWU	Female	0	Yes	Yes	9.0	Yes	No	No	No internet serv	No internet serv	No internet

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 3, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Pape
	No	Fiber optic	No	No	No	No	Yes	Yes	Month-to-month	Yes
	Yes	DSL	Yes	Yes	Yes	Yes	Yes	No	Month-to-month	Yes
	Yes	DSL	Yes	Yes	No	No	No	No	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	Month-to-month	No
	Yes	Fiber optic	No	No	No	No	Yes	No	Month-to-month	Yes
	No phone servi	DSL	No	No	No	No	No	No	Month-to-month	No
	No	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No
	Yes	Fiber optic	Yes	No	No	No	No	No	Month-to-month	Yes
	Yes	DSL	No	No	No	No	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-month	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet se	No internet se	No internet service	One year	Yes

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 3, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	Yes	Yes	Month-to-month	Yes	Mailed check	89.55	3856.75	Yes
	Yes	Yes	No	Month-to-month	Yes	Credit card (automatic)		2976.95	No
	No	No	No	Two year	No	Bank transfer (automatic)	59.75	4265.0	No
ternet service	No internet ser	No internet se	No internet service	Month-to-month	No	Mailed check	20.35		No
	No	Yes	No	Month-to-month	Yes	Credit card (automatic)	85.05	1391.15	No
	No	No	No	Month-to-month	No	Credit card (automatic)	25.1	382.8	No
	Yes	Yes	Yes	Two year	No	Bank transfer (automatic)	85.3	6129.2	No
	No	No	No	Month-to-month	Yes	Electronic check	79.8	1001.2	No
	No	No	Yes	Month-to-month	Yes	Bank transfer (automatic)	63.05	1067.05	No
	No	Yes	Yes	Month-to-month	Yes	Credit card (automatic)	102.95	2496.7	Yes

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 4, columnas 1

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customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProte
8104-OSKWT	Female	0	No	No	12.0	Yes	Yes	Fiber optic	Yes	No	No
3688-FTHLT	Female	0	No	No	16.0	Yes	Yes	DSL	No	No	No
2034-CGRHZ	Male	1	No	No	24.0	Yes	Yes	Fiber optic	No	Yes	Yes
2667-WYLWJ	Female	0	Yes	Yes	8.0	Yes	No	No	No internet serv	No internet serv	No internet
1089-HDMKP	Male	1	No	Yes	11.0	Yes	No	DSL	No	Yes	No
4570-QHXHL	Female	0	No	No	9.0	Yes	No	DSL	No	No	No
1689-MRZQR	Male	0	Yes	Yes	34.0	Yes	No	Fiber optic	No	No	No
8780-YRMTT	Female	0	No	No	66.0	No	No phone servi	DSL	Yes	Yes	Yes
0936-NQLJU	Female	0	Yes	No	41.0	Yes	No	No	No internet serv	No internet serv	No internet
4488-PSYCG	Male	0	No	No	16.0	Yes	No	No	No internet serv	No internet serv	No internet

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 4, columnas 2

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Service	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	Paper
	Yes	Fiber optic	Yes	No	No	No	No	No	Month-to-month	Yes
	Yes	DSL	No	No	No	No	No	Yes	Month-to-month	Yes
	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-month	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	One year	Yes
	No	DSL	No	Yes	No	Yes	No	No	Month-to-month	Yes
	No	DSL	No	No	No	No	No	No	Month-to-month	Yes
	No	Fiber optic	No	No	No	No	Yes	No	Month-to-month	No
	No phone servi	DSL	Yes	Yes	Yes	Yes	No	No	Two year	Yes
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Two year	No
	No	No	No internet serv	No internet serv	No internet service	No internet ser	No internet se	No internet service	Month-to-month	No

Vista previa: telco_csv_3.csv · primeras 30 filas · filas 4, columnas 3

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eProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	No	No	No	Month-to-month	Yes	Electronic check	79.8	1001.2	No
	No	No	Yes	Month-to-month	Yes	Bank transfer (automatic)	63.05	1067.05	No
	No	Yes	Yes	Month-to-month	Yes	Credit card (automatic)	102.95	2496.7	Yes
ternet service	No internet se	No internet se	No internet service	One year	Yes	Mailed check	19.75	145.0	Yes
	Yes	No	No	Month-to-month	Yes	Electronic check	1263.4012276270	601.25	No
	No	No	No	Month-to-month	Yes	Electronic check	43.75	405.7	No
	No	Yes	No	Month-to-month	No	Electronic check	78.3	2564.3	Yes
	Yes	No	No	Two year	Yes	Mailed check	47.4	3177.25	No
ternet service	No internet se	No internet se	No internet service	Two year	No	Credit card (automatic)	1458.5204738962	875.55	No
ternet service	No internet se	No internet se	No internet service	Month-to-month	No	Bank transfer (automatic)	20.25	327.45	Yes